HDDA_Project_Group2

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2026-01-21

Part 1

Singular Value Decomposition.

It is a mathematical tool used to simplify a matrix by factorizing it into three simpler matrix. The matrix will be decomposed into two orthogonal matrices, one with left singular vectors (eigenvectors of AA^T) and the other with the transpose of the right singular vectors (eigenvectors of A^TA) of the matrix and the third a diagonal matrix whose entries are single values of the matrix (in descending order).

As an example, let us perform Singular Value Decomposition on a matrix of our choice.

```r
# Define your matrix
A <- matrix(c(-1, 3, 2, 4), nrow = 2, ncol = 2)

# Calculate the transpose
A_transpose <- t(A)

# Compute AA^T
AA_transpose <- A %*% t(A)

# Find the Eigenvalues of AA^T
ev <- eigen(AA_transpose)
eigenvalues <- ev$values

# Find the right singular vectors
A_transposeA <- t(A) %*% A
ev_v <- eigen(A_transposeA)
V <- ev_v$vectors
V_transpose <- t(V)

# Compute the diagonal matrix
singular_values <- sqrt(ev_v$values)
Sigma <- diag(singular_values)

# Compute the left singular vectors
U <- A %*% V %*% solve(Sigma)

# Compute the final SVD equation
SDVeq <- U %*% Sigma %*% V_transpose


print(A)
```

```
##      [,1] [,2]
```

```
## [1,]   -1    2
## [2,]    3    4
```

```r
print(U)
```

```
##           [,1]       [,2]
## [1,] 0.2297529  0.9732490
## [2,] 0.9732490 -0.2297529
```

```r
print(Sigma)
```

```
##          [,1]     [,2]
## [1,] 5.116673 0.000000
## [2,] 0.000000 1.954395
```

```r
print(V_transpose)
```

```
##            [,1]      [,2]
## [1,]  0.5257311 0.8506508
## [2,] -0.8506508 0.5257311
```

In practice, SDV is useful in dimensionality reduction, noise reduction and matrix approximation. SVD is also used in image and text processing to compress images and enabling text-based recommendation respectively.

Correspondence Analysis

Correspondence Analysis (CA) is a technique used to explore the relationship between *two* categorical variables usually in a contingency table. A contingency table is a table with the frequency distribution of the variables under question, showing how they relate with each other. CA will explain what is actually observed in the table and also what we would expect if the two variables had no relationship whatsoever. This technique will group together all categories that behaves similarly (appear with the same categories of the other variable), and place categories that differ far apart. CA will then reduce the information in the table to low-dimensions (usually two) so that the data can be visualiased graphically.

The following code shows an example of how exactly CA is carried out.

```r
# Contingency table
N <- matrix(c(20, 10, 5,
                15, 25, 10,
                5, 10, 30),
              nrow = 3, byrow = TRUE)

rownames(N) <- c("Young", "Adult", "Senior")
colnames(N) <- c("Car", "Bus", "Bike")

# Convert counts to proportions
n <- sum(N)
P <- N / n     # correspondence matrix

# Compute row and column masses
r <- rowSums(P)    # row totals
c <- colSums(P)    # column totals

# center and standardize the matrix
Dr_inv_sqrt <- diag(1 / sqrt(r))
Dc_inv_sqrt <- diag(1 / sqrt(c))

S <- Dr_inv_sqrt %*% (P - r %*% t(c)) %*% Dc_inv_sqrt
```

```r
# Apply SVD
svd_CA <- svd(S)

U <- svd_CA$u
Sigma <- diag(svd_CA$d)
V <- svd_CA$v

# coordinates for plotting
row_coords <- Dr_inv_sqrt %*% U %*% Sigma
col_coords <- Dc_inv_sqrt %*% V %*% Sigma

row_coords
```

```
##            [,1]        [,2]         [,3]
## [1,] -0.5213698  0.29014236 1.791624e-17
## [2,] -0.2469560 -0.26323093 1.791624e-17
## [3,]  0.6799054  0.06681252 1.791624e-17
```

```r
col_coords
```

```
##            [,1]        [,2]          [,3]
## [1,] -0.5296049  0.24259838 -1.791624e-17
## [2,] -0.2012729 -0.29671715 -1.791624e-17
## [3,]  0.6720328  0.08107414 -1.791624e-17
```

This process can be applied automatically on any programming tool. We shall use an R code to show an example and give an intepretation of the results obtained.

```r
# Create a contingency table
data <- matrix(c(20, 10, 5,
                 15, 25, 10,
                 5, 10, 30),
               nrow = 3, byrow = TRUE)

rownames(data) <- c("Young", "Adult", "Senior")
colnames(data) <- c("Car", "Bus", "Bike")

# Convert to table
data_table <- as.table(data)

# Perform Correspondence Analysis
ca_result <- ca::ca(data_table)

# Eigenvalues
ca_result$eig
```

```
## NULL
```

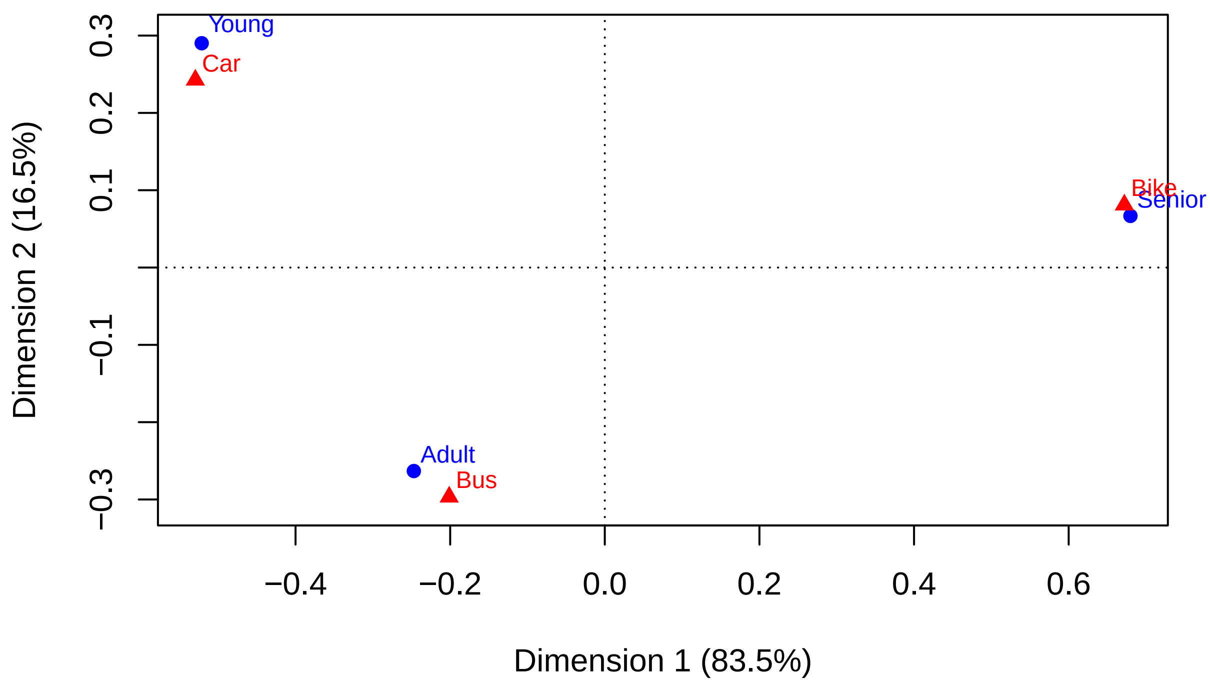
```r
# Row coordinates
ca_result$rowcoord
```

```
##              Dim1       Dim2
## Young  -1.0291264  1.2865397
## Adult  -0.4874638 -1.1672099
## Senior  1.3420582  0.2962579
```

```r
# Column coordinates
ca_result$colcoord
```

```
##            Dim1       Dim2
## Car  -1.0453816  1.0757218
## Bus  -0.3972904 -1.3156934
## Bike  1.3265185  0.3594963
```

```r
# Plot CA results
plot(ca_result)
```



Young individuals are strongly associated with cars, while adults are associated with buses, as shown by their close proximity on the plot. Seniors are closely associated with bikes and are clearly separated from the other groups along Dimension 1, which captures most of the variation (83.5%). Categories far from the origin (e.g., Young–Car and Senior–Bike) represent the strongest and most distinctive relationships.

In practice, CA is widely used in market research, social sciences, and survey analysis to explore relationships between categorical variables.

Multiple Correspondence analysis

Multiple Correspondence Analysis (MCA)is an extension of CA. It is used when we have more than two categorical variables (usually from a questionnaire or survey). First, the technique forms the Indicator Matrix which is the answers in binary format (0 or 1). Each possible answer becomes a separate column, that is if a question has 3 possible answers (A, B, C), it becomes three columns. MCA then looks for patterns in these answers. Answers that are often chosen by the same people are considered similar and vise-versa. MCA will then summarise all the relationships into low-dimensional matrix (two or three), which represents the strongest pattern in the data.

Key Concepts of MCA

Inertia: This is the term for "variance" in MCA. It tells us how spread out the categories are.

Dimensionality Reduction: It condenses dozens of survey questions into 2 or 3 "latent" factors (e.g., Urban vs. rural lifestyles).

Practical Use of MCA

Identifying profiles of respondents. For example, discovering that people who buy a certain Product X also tend to be aged 18-25, live in cities, and enjoy sports.

```r
# Create a data frame
data <- data.frame(
  Age    = c("Young", "Young", "Old", "Old"),
  City   = c("Yes", "No", "Yes", "No"),
  Sports = c("Yes", "Yes", "No", "No"))

# Indicator matrix
X <- model.matrix(~ Age + City + Sports - 1, data = data)

# Convert to proportions
n <- sum(X)
P <- X / n

# Compute masses
r <- rowSums(P)
c <- colSums(P)

# Standardise
Dr_inv_sqrt <- diag(1 / sqrt(r))
Dc_inv_sqrt <- diag(1 / sqrt(c))

S <- Dr_inv_sqrt %*% (P - r %*% t(c)) %*% Dc_inv_sqrt

# Apply SVD
svd_MCA <- svd(S)

U <- svd_MCA$u
Sigma <- diag(svd_MCA$d)
V <- svd_MCA$v

# category coordinates
category_coords <- Dc_inv_sqrt %*% V %*% Sigma
category_coords
```

```
##            [,1]       [,2]          [,3]          [,4]
## [1,]  1.3660254 -0.3660254 -7.376094e-17  2.752424e-19
## [2,] -0.7886751 -0.2113249 -7.900087e-17 -7.473765e-18
## [3,]  0.2113249  0.7886751 -7.376094e-17  2.752424e-19
## [4,] -0.7886751 -0.2113249 -6.852101e-17  8.024250e-18
```

Lets now apply the process using R functions and visualise the results.

```r
# Create a data frame
data <- data.frame(
  Age    = c("Young", "Young", "Old", "Old"),
  City   = c("Yes", "No", "Yes", "No"),
  Sports = c("Yes", "Yes", "No", "No"))

# Perform MCA
mca_result <- MCA(data, graph = FALSE)

# Eigenvalues
mca_result$eig
```

```
##         eigenvalue percentage of variance cumulative percentage of variance
## dim 1 6.666667e-01           6.666667e+01                          66.66667
## dim 2 3.333333e-01           3.333333e+01                         100.00000
## dim 3 7.912901e-34           7.912901e-32                         100.00000
```

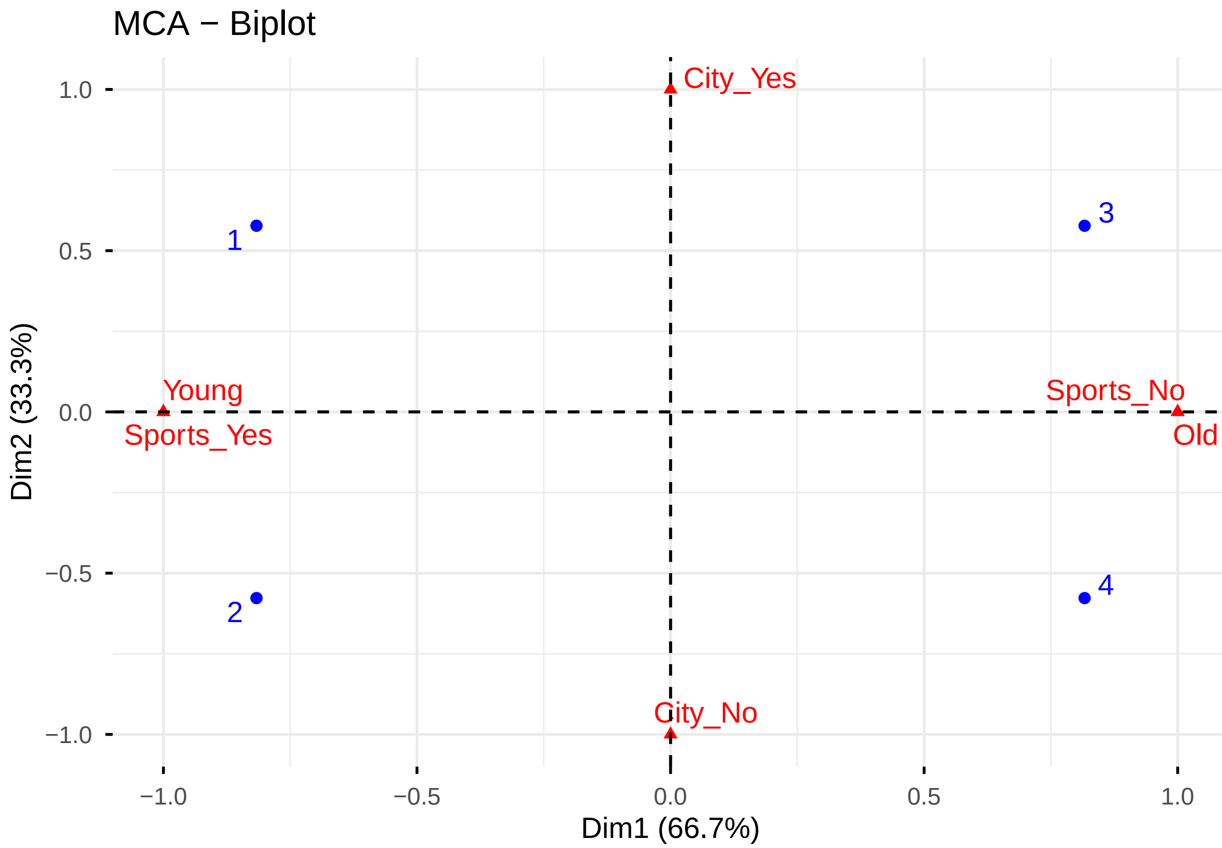
```r
# Individual coordinates
mca_result$ind$coord
```

```
##        Dim 1      Dim 2        Dim 3
## 1 -0.8164966  0.5773503 9.936144e-34
## 2 -0.8164966 -0.5773503 5.147916e-34
## 3  0.8164966  0.5773503 5.147916e-34
## 4  0.8164966 -0.5773503 9.936144e-34
```

```r
# Variable categories coordinates
mca_result$var$coord
```

```
##                    Dim 1         Dim 2         Dim 3
## Old         1.000000e+00 -3.140185e-16  1.063295e-33
## Young      -1.000000e+00  3.925231e-16 -8.701438e-34
## City_No    -3.330669e-16 -1.000000e+00  0.000000e+00
## City_Yes    3.330669e-16  1.000000e+00  0.000000e+00
## Sports_No   1.000000e+00 -3.140185e-16 -9.667195e-34
## Sports_Yes -1.000000e+00  3.140185e-16  9.667195e-34
```

```r
# Plot MCA results
fviz_mca_biplot(mca_result, repel = TRUE)
```



Dimension 1 (66.7%) contrasts young, sports-oriented individuals with older, non-sports individuals, as shown by Young + Sports_Yes on the left and Old + Sports_No on the right. Dimension 2 (33.3%) mainly separates city residents from non-city residents, with City_Yes at the top and City_No at the bottom. Respondents located near these category points share similar profiles, while categories far from the origin represent the most distinctive characteristics.

In practice, MCA is commonly used in survey analysis, sociology, psychology, and market segmentation to uncover hidden structures in categorical data.

Part II

```r
dataset <- read.csv("HDDAdataexam26.csv", sep = ";", dec = ",", header = TRUE, row.names = NULL)
head(dataset)
```

```
##   Phone SocialNetworks Happiness Walk InstagramRatio
## 1   360            360         6   12      1.2677165
## 2   180            130         6    3      0.5420827
## 3   250            120         7    8      0.7500000
## 4   120             80         8    7      1.7964072
## 5   190              1         3   10      0.8465608
## 6   120             60         7   10      1.0371820
```

```r
sum(is.na(dataset))
```

```
## [1] 0
```

```r
sum(duplicated(dataset))
```

```
## [1] 1
```

```r
dataset <- dataset[!duplicated(dataset), ]
```

```r
sum(duplicated(dataset))
```

```
## [1] 0
```

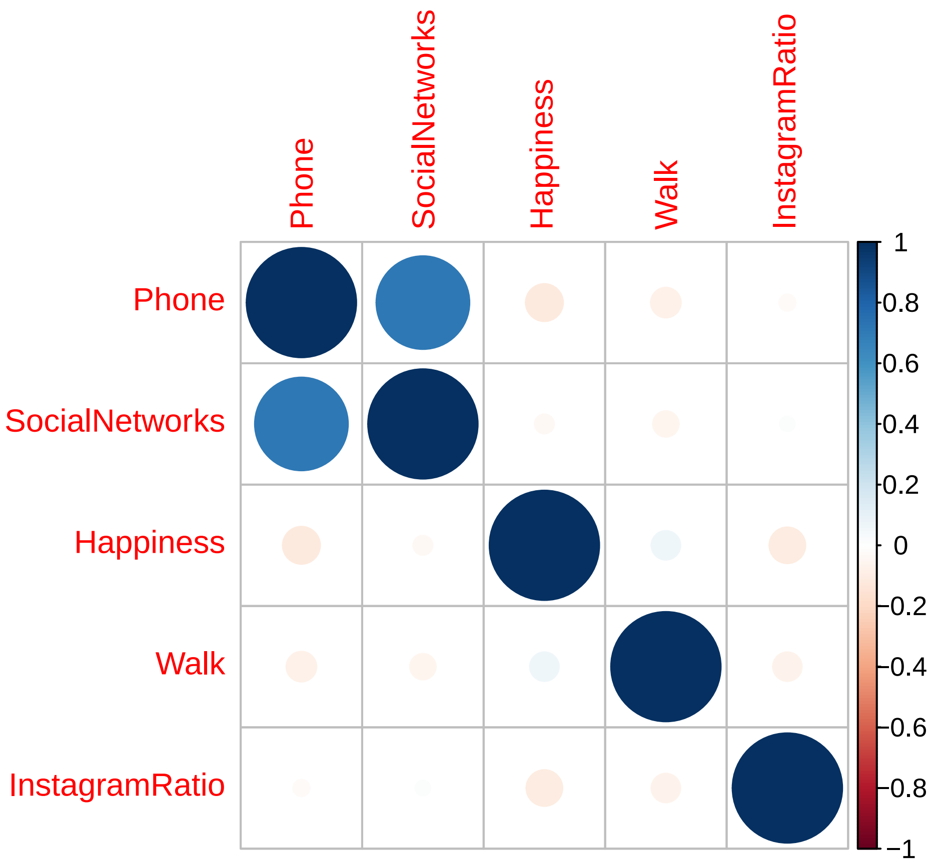
```r
summary(dataset)
```

```
       Phone          SocialNetworks      Happiness           Walk
   Min.   :   1.00   Min.   :   1.00   Min.   : 1.000   Min.   : 1.000
   1st Qu.:   9.75   1st Qu.:  30.00   1st Qu.: 7.000   1st Qu.: 4.000
   Median : 120.00   Median :  60.00   Median : 8.000   Median : 5.000
   Mean   : 125.89   Mean   :  80.41   Mean   : 7.462   Mean   : 6.221
   3rd Qu.: 180.00   3rd Qu.: 120.00   3rd Qu.: 8.000   3rd Qu.: 8.125
   Max.   :1440.00   Max.   :1440.00   Max.   :10.000   Max.   :20.000
   InstagramRatio
   Min.   : 0.0000
   1st Qu.: 0.5799
   Median : 0.9621
   Mean   : 1.1787
   3rd Qu.: 1.2456
   Max.   :51.3333
```

```r
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```r
corrplot(cor(dataset))
```




```r
res.pca <- PCA(dataset, graph = FALSE, scale.unit = FALSE)
print(res.pca)
```

```
## **Results for the Principal Component Analysis (PCA)**
## The analysis was performed on 368 individuals, described by 5 variables
## *The results are available in the following objects:
##
##    name               description
## 1  "$eig"             "eigenvalues"
## 2  "$var"             "results for the variables"
## 3  "$var$coord"       "coord. for the variables"
## 4  "$var$cor"         "correlations variables - dimensions"
## 5  "$var$cos2"        "cos2 for the variables"
## 6  "$var$contrib"     "contributions of the variables"
## 7  "$ind"             "results for the individuals"
## 8  "$ind$coord"       "coord. for the individuals"
## 9  "$ind$cos2"        "cos2 for the individuals"
## 10 "$ind$contrib"     "contributions of the individuals"
## 11 "$call"            "summary statistics"
## 12 "$call$centre"     "mean of the variables"
## 13 "$call$ecart.type" "standard error of the variables"
## 14 "$call$row.w"      "weights for the individuals"
## 15 "$call$col.w"      "weights for the variables"
```

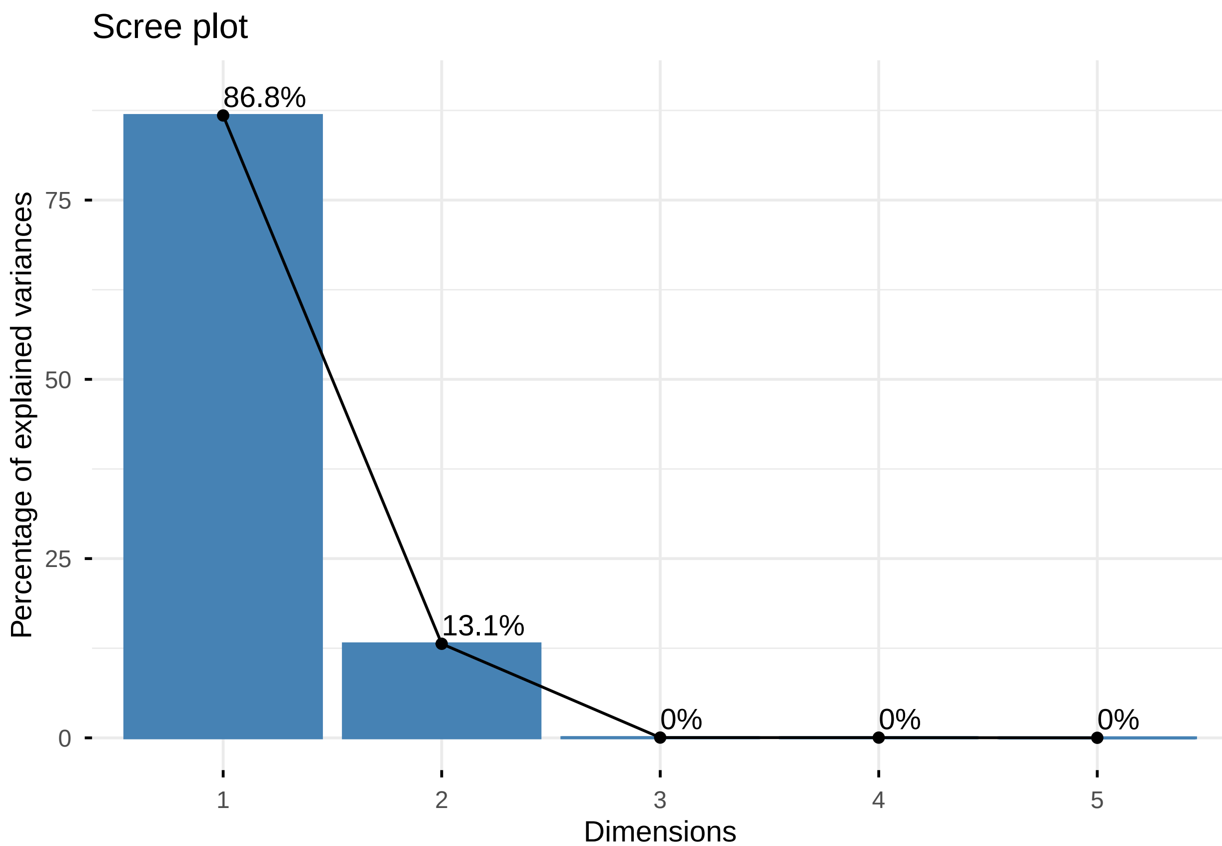
Eigen values

```
##           eigenvalue percentage of variance cumulative percentage of variance
```

```
## comp 1 24790.626106          86.801507272                       86.80151
## comp 2  3746.359682          13.117444706                       99.91895
## comp 3    11.061297           0.038729850                       99.95768
## comp 4     9.559954           0.033473071                       99.99115
## comp 5     2.526173           0.008845101                      100.00000
```

```
fviz_eig(res.pca, addlabels = TRUE, ylim = c(0, 90))
```

```
## Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
## Ignoring empty aesthetic: `width`.
```



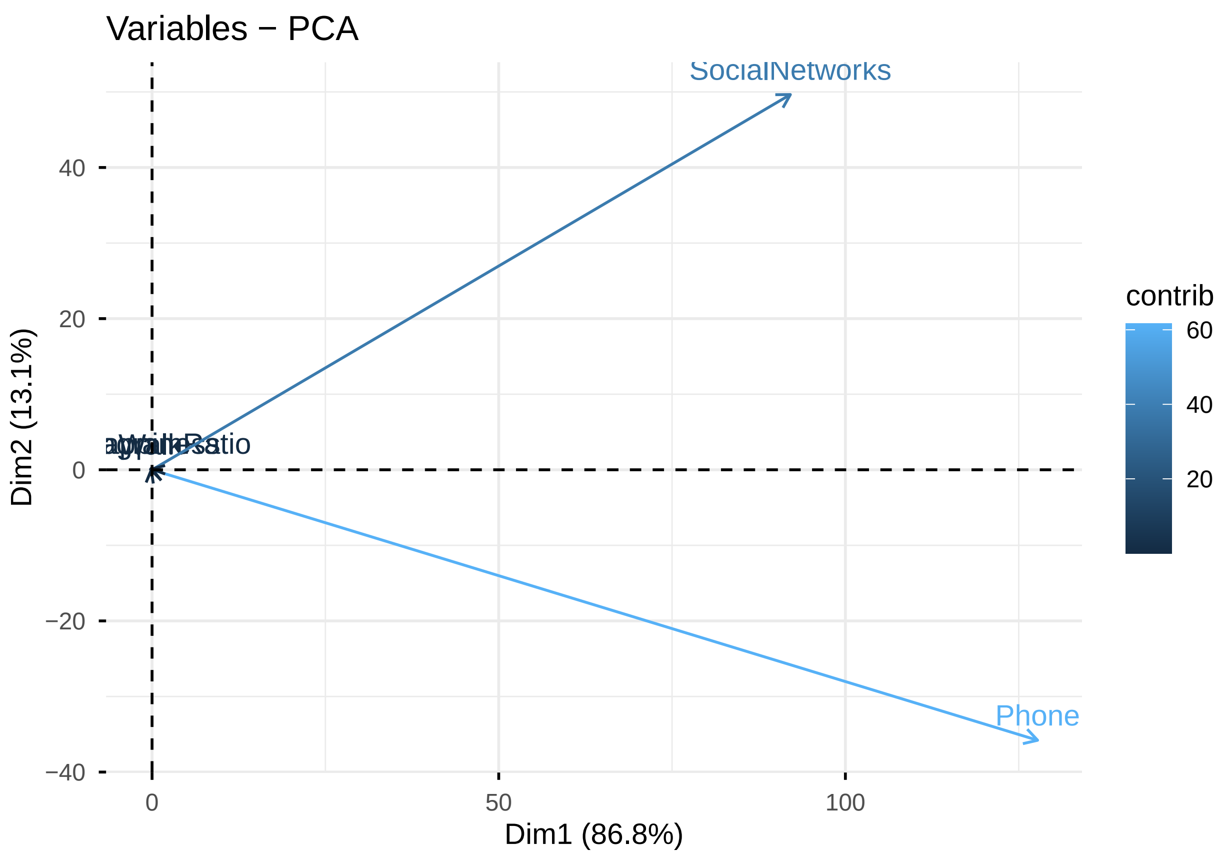
The variables

```
## $coord
##                         Dim.1        Dim.2        Dim.3        Dim.4        Dim.5
## Phone            127.72405150 -35.79147563  0.001029120  0.007421044  0.003833565
## SocialNetworks    92.07126143  49.65148148  0.004575761 -0.004623555 -0.002910348
## Happiness         -0.14590388   0.16735516  0.245249108 -0.098437042  1.584252129
## Walk              -0.23169911   0.05870641  2.469310001  2.068640434 -0.053342058
## InstagramRatio    -0.02491867   0.16996252 -2.214415447  2.295847152  0.115971451
##
## $cor
##                        Dim.1       Dim.2         Dim.3         Dim.4
## Phone             0.96290774 -0.26983085  7.758503e-06  5.594702e-05
## SocialNetworks    0.88017304  0.47465294  4.374287e-05 -4.419977e-05
## Happiness        -0.08998551  0.10321548  1.512562e-01 -6.071057e-02
## Walk             -0.07172027  0.01817201  7.643516e-01  6.403281e-01
```

```
## InstagramRatio -0.00779565  0.05317172 -6.927661e-01  7.182415e-01
##                       Dim.5
## Phone            2.890113e-05
## SocialNetworks -2.782203e-05
## Happiness        9.770798e-01
## Walk            -1.651153e-02
## InstagramRatio  3.628095e-02
##
## $cos2
##                       Dim.1        Dim.2        Dim.3        Dim.4        Dim.5
## Phone          9.271913e-01 0.0728086852 6.019437e-11 3.130070e-09 8.352754e-10
## SocialNetworks 7.747046e-01 0.2252954175 1.913439e-09 1.953620e-09 7.740656e-10
## Happiness      8.097392e-03 0.0106534358 2.287844e-02 3.685773e-03 9.546850e-01
## Walk           5.143797e-03 0.0003302221 5.842333e-01 4.100201e-01 2.726306e-04
## InstagramRatio 6.077216e-05 0.0028272314 4.799249e-01 5.158708e-01 1.316307e-03
##
## $contrib
##                       Dim.1        Dim.2        Dim.3        Dim.4        Dim.5
## Phone          6.580485e+01 3.419399e+01 9.574715e-06 5.760686e-04 5.817584e-04
## SocialNetworks 3.419485e+01 6.580440e+01 1.892869e-04 2.236126e-04 3.352948e-04
## Happiness      8.587094e-05 7.475990e-04 5.437620e-01 1.013588e-01 9.935405e+01
## Walk           2.165515e-04 9.199444e-05 5.512457e+01 4.476249e+01 1.126358e-01
## InstagramRatio 2.504737e-06 7.710754e-04 4.433147e+01 5.513535e+01 5.324013e-01
```

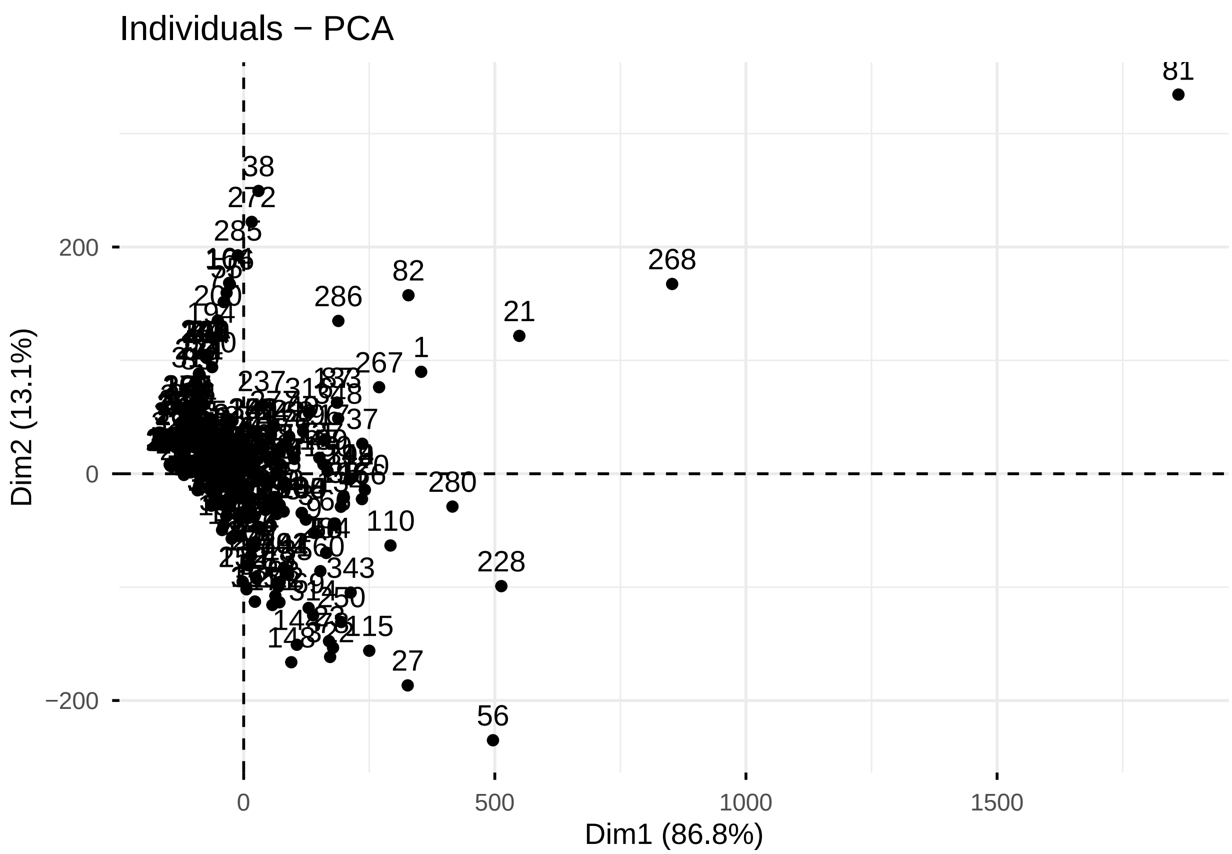
```
fviz_pca_var(res.pca, col.var = "contrib")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## i The deprecated feature was likely used in the ggpubr package.
##   Please report the issue at <https://github.com/kassambara/ggpubr/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



The invividuals

```
fviz_pca_ind(res.pca)
```



1) Construct the following variables and add them to the dataframe 'dataset'- 'LowInstagram' is a dummy taking the value 1 if 'InstagramRatio' is less than 0.5, and zero otherwise;- 'HighInstagram' is a dummy taking the value 1 if 'InstagramRatio' is greater than 2, and zero otherwise;

```
dataset$LowInstagram  <- ifelse(dataset$InstagramRatio < 0.5, 1, 0)
dataset$HighInstagram <- ifelse(dataset$InstagramRatio > 2, 1, 0)
```

2) We would like to investigate the smartphone addiction of AIMS students. Do a multiple regression with 'Phone' as dependent variable (Y) and the following independent (predictors) variables: 'SocialNetworks', 'Happiness', 'Walk', 'LowInstagram', and 'HighInstagram'.

```
model_full <- lm(Phone ~ SocialNetworks + Happiness + Walk +
             LowInstagram + HighInstagram, data = dataset)
summary(model_full)
```

```
##
```

```
## Call:
## lm(formula = Phone ~ SocialNetworks + Happiness + Walk + LowInstagram +
##     HighInstagram, data = dataset)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -296.75  -43.55   -4.38   38.57  422.64
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    111.88248   25.24504   4.432 1.24e-05 ***
## SocialNetworks   0.91075    0.04555  19.996  < 2e-16 ***
## Happiness       -7.39517    2.97650  -2.485   0.0134 *
## Walk            -0.95744    1.48481  -0.645   0.5194
## LowInstagram    17.90758   11.60507   1.543   0.1237
## HighInstagram  -34.77905   19.97274  -1.741   0.0825 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 91.17 on 362 degrees of freedom
## Multiple R-squared:  0.5352, Adjusted R-squared:  0.5288
## F-statistic: 83.38 on 5 and 362 DF,  p-value: < 2.2e-16
```

```r
anova(model_full)
```

```
## Analysis of Variance Table
##
## Response: Phone
##                 Df  Sum Sq Mean Sq  F value  Pr(>F)
## SocialNetworks   1 3351385 3351385 403.1641 < 2e-16 ***
## Happiness        1   55719   55719   6.7029 0.01001 *
## Walk             1    5288    5288   0.6361 0.42564
## LowInstagram     1   27964   27964   3.3640 0.06746 .
## HighInstagram    1   25206   25206   3.0322 0.08248 .
## Residuals      362 3009200    8313
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

3) Is the regression model in 2) useful in predicting 'Phone' time?.

ANSWER: The p-value of the global Fisher test is smaller than 2.2e-16, which is far below the 5% significance level. Therefore, we reject the null hypothesis that all regression coefficients are zero. This shows that the linear regression model is statistically significant and useful for predicting phone usage time among AIMS students.

4) In the regression of 2), does 'SocialNetworks' significantly influence 'Phone'?

```r
summary(model_full)$coefficients["SocialNetworks", ]
```

```
##     Estimate   Std. Error      t value     Pr(>|t|)
## 9.107468e-01 4.554649e-02 1.999598e+01 1.862844e-60
```

ANSWER: The variable SocialNetworks has a positive and highly significant effect on phone usage. Each additional minute spent on social networks increases phone time by approximately 0.91 minutes. The extremely small p-value (1.24e-60) confirms that this effect is statistically significant

5) According to the regression in 2), does 'Happiness' help reduce the time AIMS students spend on their phone?

```
summary(model_full)$coefficients["Happiness", ]
```

```
##    Estimate  Std. Error     t value    Pr(>|t|)
## -7.39516822  2.97650428 -2.48451456  0.01342415
```

ANSWER: The variable Happiness has a negative and statistically significant effect on phone usage. Each additional point of happiness reduces the daily phone time by approximately 7.4 minutes. The p-value (0.0133) confirms that this effect is significant at the 5% level.

6) Consider the regression output in 2) and two AIMS students A and B who provide exactly the same answers (120; 4; 15) for the variables 'SocialNetworks', 'Happiness' and 'Walk' respectively. Student A has an 'InstagramRatio' of 2.1 and Student B has a 'InstagramRatio' of 0.4. Give the predicted phone time for students A and B, respectively, denoted Phone(A) and Phone(B).

```
studentA <- data.frame(SocialNetworks=120, Happiness=4, Walk=15,
                LowInstagram=0, HighInstagram=1)

studentB <- data.frame(SocialNetworks=120, Happiness=4, Walk=15,
                LowInstagram=1, HighInstagram=0)

PhoneA <- predict.lm(model_full, newdata = studentA)
PhoneB <- predict.lm(model_full, newdata = studentB)

PhoneA
```

```
##        1
## 142.4508
```

```
PhoneB
```

```
##        1
## 195.1374
```

Interpretation: Using the linear regression model, the predicted phone time for student A is 142.4649, while for student B it is 195.1332 . The difference comes from the Instagram indicators, since student A has a high Instagram ratio whereas student B has a low Instagram ratio.

7) Do a backward stepwise regression until you only have significant predictors left at the 5% significance level (it means delete one by one non-significant predictors beginning by the most non-significant).

```r
model_full <- lm(
  Phone ~ SocialNetworks + Happiness + Walk + LowInstagram + HighInstagram,
  data = dataset)

backward_model1 <- step(model_full, direction = "backward")
```

```
## Start:  AIC=3327.35
## Phone ~ SocialNetworks + Happiness + Walk + LowInstagram + HighInstagram
##
##                  Df Sum of Sq     RSS    AIC
## - Walk            1      3456 3012656 3325.8
## <none>                        3009200 3327.3
## - LowInstagram    1     19793 3028993 3327.8
## - HighInstagram   1     25206 3034406 3328.4
## - Happiness       1     51313 3060513 3331.6
## - SocialNetworks  1   3323748 6332948 3599.2
##
## Step:  AIC=3325.77
## Phone ~ SocialNetworks + Happiness + LowInstagram + HighInstagram
##
##                  Df Sum of Sq     RSS    AIC
## <none>                        3012656 3325.8
## - LowInstagram    1     21788 3034444 3326.4
## - HighInstagram   1     24808 3037465 3326.8
## - Happiness       1     52861 3065518 3330.2
## - SocialNetworks  1   3344855 6357511 3598.6
```

```r
summary(backward_model1)
```

```
##
## Call:
## lm(formula = Phone ~ SocialNetworks + Happiness + LowInstagram +
##     HighInstagram, data = dataset)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -298.71  -43.58   -3.88   36.40  425.41
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    106.35368   23.72529   4.483  9.9e-06 ***
## SocialNetworks   0.91232    0.04544  20.076  < 2e-16 ***
## Happiness       -7.49562    2.97003  -2.524   0.0120 *
## LowInstagram    18.68608   11.53281   1.620   0.1060
## HighInstagram  -34.49512   19.95181  -1.729   0.0847 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 91.1 on 363 degrees of freedom
```

```
## Multiple R-squared:  0.5347, Adjusted R-squared:  0.5296
## F-statistic: 104.3 on 4 and 363 DF,  p-value: < 2.2e-16
```

Intepretation:

After implementing the backward stepwise selection using the step() function, we notice that the algorithm did not remove all non-significant variables. Although the p-value of the model asserts the significance of model, we still have to explore other models that can be obtained using the backward stepwise selection. We try another model which does not include 'LowInstagram' as it has the highest p-value from the previous results.

```
model_noLowInstagram <- lm(
  Phone ~ SocialNetworks + Happiness + HighInstagram,
  data = dataset)

backward_model2 <- step(model_noLowInstagram, direction = "backward")
```

```
## Start:  AIC=3326.42
## Phone ~ SocialNetworks + Happiness + HighInstagram
##
##                  Df Sum of Sq     RSS    AIC
## <none>                        3034444 3326.4
## - HighInstagram   1     33214 3067658 3328.4
## - Happiness       1     63523 3097967 3332.0
## - SocialNetworks  1   3340108 6374552 3597.6
```

```
summary(backward_model2)
```

```
##
## Call:
## lm(formula = Phone ~ SocialNetworks + Happiness + HighInstagram,
##     data = dataset)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -304.10  -46.25   -4.21   37.33  418.67
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)    115.80898   23.04767   5.025 7.92e-07 ***
## SocialNetworks   0.91163    0.04554  20.017  < 2e-16 ***
## Happiness       -8.14230    2.94966  -2.760  0.00606 **
## HighInstagram  -39.44311   19.76068  -1.996  0.04667 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 91.3 on 364 degrees of freedom
## Multiple R-squared:  0.5313, Adjusted R-squared:  0.5275
## F-statistic: 137.6 on 3 and 364 DF,  p-value: < 2.2e-16
```

Intepretation:

All variables are significant after removing 'LowInstagram'. We also notice that the difference in the adjusted-rsquared of the two models obtained using backward stewise selection isvery low (0.0021) so we can select the simpler model(backward_model2 with less variables but more significant variables). In addition, the second model seems to reduce the AIC which is good.

8) Choose the model among those you obtained in 2) and 7) that is the most useful one according to the significance of the predictors variables and adjusted coefficient of determination to predict 'Phone' time.

```
adjR2_model1 <- summary(model_full)$adj.r.squared
adjR2_model2 <- summary(backward_model2)$adj.r.squared

adjR2_model1
```

```
## [1] 0.5288223
```

```
adjR2_model2
```

```
## [1] 0.5274802
```

Interpretation:

The model obtained in 2) (model_full) has a smaller adjusted-rsquared that the model obtained in 7) (backward_model2) but the difference is very small (0.0013421). However 'Model_full' has more variables than 'backward_model2', majority of which are not significant. We therefore decide that that the 'backward_model2' is the most useful as it is simpler and has an okay adjusted-rsquared.

9) Using the best model chosen in 8), give the predicted time an AIMS student spends watching his/her phone per day for a student who spends 30 minutes on social networks per day, has a happiness score of 8, walks 2 kilometers a day and has an Instagram ratio of 2.1.

```
# By using the best model backward_model2
student1 <- data.frame(
  SocialNetworks = 30,
  Happiness = 8,
  HighInstagram = ifelse(2.1 > 2, 1, 0)) # HighInstagram = 1

Phone1 <- predict.lm(backward_model2, newdata = student1)
Phone1
```

```
##        1
## 38.57649
```

Intepretation:

An AIMS student who spends 30 minutes on social networks per day, has a happiness score of 8,walks 2 kilometers a day and has an Instagram ratio of 2.1 is predicted to spend roughly 38.58minutes on their phone.

10) Using the best model chosen in 8), give the predicted time an AIMS student spends watching his/her phone per day for a student who spends 30 minutes on social networks per day, has a happiness score of 8, walks 2 kilometers a day and has an Instagram ratio of 0.4.

```r
# By using the best model backward_model2
student2 <- data.frame(
  SocialNetworks = 30,
  Happiness = 8,
  HighInstagram = ifelse(0.4 > 2, 1, 0)) # HighInstagram = 0

Phone2 <- predict.lm(backward_model2, newdata = student2)
Phone2
```

```
##       1
## 78.0196
```

Intepretantion

An AIMS student who spends 30 minutes on social networks per day, has a happiness score of 8, walks 2 kilometers a day and has an Instagram ratio of 0.4 is predicted to spend approximately 78.02 minutes on their phone.

Conclusion

The final model shows that being in the 'highInstagram' group is associated with a significant decrease in daily Phone time, suggesting that 'influencer-type' profiles may have more curated or efficient usage patterns compared to others users.

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